



AI RECOMMENDATION SYSTEM

Virtual Internship Project 3: Knowledge-Based Filtering Architecture

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Program: DecodeLabs Internship (Week 3)

PROJECT OVERVIEW

The Domain Challenge

Building a logical personalization pipeline that segments cross-domain catalogs (Movies, Books, Games) efficiently according to explicit user preference input.

Our System Approach

A multi-layered conditional decision architecture that dynamically maps structural content sub-types to predefined preference matrices using dynamic input normalization.

MULTI-DOMAIN CATALOG



Movies

Segmented dynamically into
Action, Comedy, and Horror tiers.



Books

Classified under educational and
leisure paths: Science, History, and
Fiction.



Games

Mapped instantly for active users
across Racing, Adventure, and
Sports vectors.

SYSTEM ARCHITECTURE STRATEGY

- ↓ ⓘ **Hierarchical Filtering:** Implements nested logical structures to isolate main classes before drilling down to individual sub-genres.
- A **Input Normalization:** Utilizes explicit Python `.lower()` mutations on user strings to achieve robust validation against system rule-sets.
- 🛡️ **Exception Mitigation:** Integrated catch-all conditional fallbacks to flag invalid inputs structural categories gracefully.

CONTENT MATRIX STRATEGY

Domain Class	Sub-Genre Vector	Top Pick Deployments
Movies	Action Comedy Horror	John Wick, Home Alone, The Conjuring
Books	Science History Fiction	Cosmos, Sapiens, Harry Potter
Games	Racing Adventure Sports	Forza Horizon, GTA V, EA FC 26

INTERACTIVE TERMINAL OUTPUT

DecodeLabs Recommendation Engine Console v3.0

Categories:

1. Movies
2. Books
3. Games

Choose a category: **movies**

Movie Types:

- Action
- Comedy
- Horror

Choose movie type: **action**

Recommended Movies:

1. John Wick
2. Mission Impossible
3. Mad Max

Deterministic Intent Resolution

The console output showcases nested lookup execution.

Normalization correctly resolves case variations, verifying the structural accuracy of user recommendations instantly.

THANK YOU

DecodeLabs Virtual Internship | Week 3

Completed

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